

Label-free cold start for active learning beats a supervised evolutionary envelope

— if the envelope is honest: a circularity audit

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Abstract

The first labeled set L_0 decides how an active learning loop starts — and in short-text classification with hundreds of imbalanced classes, it decides a lot: we measure a 6.4-point accuracy spread at $|L_0| = 100$ caused by the random draw alone (47 sizes $\times$ 30 repetitions, independently re-executed with ≤ 0.7 -point divergence). To estimate how much deliberate composition can yield, a genetic algorithm optimizes L_0 composition under four objectives, revealing a wide envelope (up to 25.9 points between best and worst individuals at $|L_0| = 100$) that narrows with size. We then show that DRI-SL — a deterministic, label-free, linear-cost heuristic combining semantic density (embedding-cluster representativeness) with lexical variety — *outperforms the best individual found by the supervised optimizer* at every size from 100 to 5,000 (e.g., 67.4% vs. 62.2% accuracy at $|L_0| = 1,000$). Finally, a methodological audit shows the standard protocol of reporting evolutionary envelopes on the same partition used as fitness inflates them — by 6.3 points of macro-F1 at $|L_0| = 500$ in our re-execution with an anti-circularity protocol — which means published evolutionary bounds for subset selection are optimistic and our comparison is conservative. Composition, not just quantity, governs the cold-start regime; it can be captured without any labels; and envelopes must be reported on untouched partitions.

Keywords: active learning; cold start; instance selection; genetic algorithms; evaluation methodology; short text.

1 Introduction

Pool-based active learning assumes a model good enough to guide selection — which presupposes an initial labeled set L_0 . In the cold-start regime the loop’s later efficiency is hostage to this first draw (Bayer and Reuter, 2024; Attenberg and Provost, 2010). Three questions structure this paper: (RQ1) how much does the *composition* of L_0 matter, at fixed size? (RQ2) how much headroom does deliberate composition have over random draws — an envelope we estimate with a supervised evolutionary optimizer? (RQ3) can a label-free constructive heuristic capture that headroom? A fourth, methodological question emerged from answering the second: (RQ4) are evolutionary envelopes, as usually reported, even honest?

We answer on a hard instance of the problem: 250k short product descriptions (4–50 characters, abbreviations) in Portuguese, 621 imbalanced classes, where class coverage — not instance count — dominates early performance.

2 Related work

Cold-start-aware AL uses pretrained representations to pick informative seeds: ALPS exploits language-model surprisal (Yuan et al., 2020); BERT-era studies adapt AL to fine-tuning (Ein-Dor et al., 2020; Griefhaber et al., 2020); DEUCE combines textual and class-prediction diversity (Guo et al., 2025); LLM-based selectors push the idea further (Bayer and Reuter, 2024). Density/cluster-based selection is classical (Settles and Craven, 2008; Nguyen and Smeulders, 2004; Dasgupta and Hsu, 2008), as are batch-diversity concerns (Hoi et al., 2006). Evolutionary algorithms (Holland, 1975; Goldberg, 1989) are widely used for instance and subset selection; we are not aware of prior work auditing the *evaluation circularity* of their reported envelopes in this setting — fitness measured and reported on the same partition — although the underlying overfitting-to-selection phenomenon is well known in model selection.

3 Setting

Classifier: PVBIn, a per-class prototype model over binary term vectors, cheap enough for tens of thousands of retrainings (Darú, 2024). Data: deduplicated short product descriptions, 621 classes, heavy long tail; evaluation on a fixed untouched test set (20,000 instances in the re-execution protocol). Metrics: global accuracy and macro-F1, the latter primary under imbalance (Sokolova and Lapalme, 2009).

4 RQ1: sensitivity to the draw

Across 47 sizes (10 to 200,000) with 30 random repetitions each: mean accuracy grows from 6.7% ($|L_0| = 10$) through 24.7% (100), 55.9% (1,000) and 76.9% (10,000), saturating near 89.1% (200,000); macro-F1 lags (0.4% $\rightarrow$ 3.4% $\rightarrow$ 17.8% $\rightarrow$ 43.3% $\rightarrow$ 70.4%), reflecting the cost of covering minority classes. At $|L_0| = 100$, accuracy across repetitions spans **21.7–28.1%** — a **6.4-point** spread caused exclusively by the draw. The mechanism is coverage: at $|L_0| = 10$ a typical draw covers 9 of 621 classes; at 1,000, about 255; macro-F1 tracks the coverage curve closely. Good and bad L_0 of the same size *exist*; the question becomes whether they can be found without labels.

Independent re-execution. The full study was re-executed in a clean-room reimplementaion with a stricter protocol (deduplication *before* partitioning; fixed test set), on a reduced grid (15 log-spaced sizes $\times$ 10 repetitions). Agreement is within 0.7 points at every size, always in the expected direction (the deduplicated protocol is slightly harder), with implementation fidelity additionally verified by exact equality of score matrices.

5 RQ2: the evolutionary envelope

A genetic algorithm over subset membership (population 50, 100 generations, elitism; full configuration and per-cell checkpoints in the released legacy artifacts) optimizes L_0 composition for four objectives (max/min $\times$ accuracy/macro-F1) at ten sizes. The envelope is wide where it matters: at $|L_0| = 100$, best-minus-worst final individuals span **25.9 accuracy points**; evolutionary gain over the first generation (e.g., +10.1 points at 100, +11.7 at 50) confirms the optimizer exploits real structure, not noise. The envelope narrows monotonically with size (at 30,000: +0.8), consistent with RQ1: composition is a small-regime lever.

6 RQ3: a label-free heuristic vs. the supervised optimizer

DRI-SL (Density-Representative Instance Selection with Lexical variety) is deterministic given the embedding model and uses no labels: embed the pool with dense text embeddings, cluster (k -means), allocate the budget proportionally to cluster mass, and fill each cluster’s quota greedily preferring instances that add unseen lexical tokens — linear cost after clustering. Table 1 compares it with the GA envelope.

Table 1: Cold start: DRI-SL vs. GA envelope (final generation), PVBIn. GA values inherit the original protocol (fitness partition = report partition) and are therefore optimistic — see Section 7.

$ L_0 $	DRI-SL		GA (best)		GA (worst)	
	Acc	M-F1	Acc	M-F1	Acc	M-F1
100	41.2	6.9	38.8	6.5	5.8	1.8
500	59.2	18.5	56.0	18.0	36.2	6.5
1,000	67.4	25.8	62.2	24.7	49.3	12.6
2,500	73.4	36.5	69.3	32.6	63.9	24.1
5,000	76.9	44.1	74.1	40.1	70.8	31.9

A constructive heuristic with zero label access beats an optimizer that consumed thousands of supervised evaluations, at every size — strong evidence that semantic density plus lexical variety capture precisely the structure that makes an L_0 informative (class coverage through cluster coverage; feature coverage through lexical novelty). At 100 the margin over the GA’s best is 2.5 points; over the GA’s *worst*, 35 points — the cost of a bad draw that DRI-SL eliminates deterministically.

7 RQ4: the circularity audit

The GA numbers above inherit the field’s standard protocol: fitness evaluated on a partition, and the *same* partition reported as the result. Our re-execution applies an anti-circularity protocol — fitness on a selection partition, final individuals *re-evaluated once* on the untouched test set. The mechanism reproduces (best individual gains +5.2 accuracy points at $|L_0| = 50$), but the honest re-evaluation reveals the inflation: for macro-F1 maximization at $|L_0| = 500$, final fitness reads 19.4% while the same individual measures **13.1%** on untouched data — **6.3 points of overfitting to the fitness partition** (for accuracy, ≈ 1 point). Two consequences: (i) published

evolutionary envelopes for subset selection should be read as optimistic bounds unless the protocol states otherwise; (ii) our central comparison is *conservative* — DRI-SL beats an inflated opponent.

8 Conclusion

Composition of the initial labeled set is a first-order factor in cold-start active learning for long-tailed short text (6.4-point spread from the draw alone at $|L_0| = 100$); a supervised evolutionary optimizer confirms wide headroom; a deterministic label-free heuristic captures that headroom, beating the optimizer’s best individual up to $|L_0| = 5,000$; and evolutionary envelopes must be reported on untouched partitions — ours was inflated by up to 6.3 macro-F1 points before correction. In the continuous (post-cold-start) regime, companion work shows the lexical-variety component should be dropped in favor of plain prediction-stratified sampling; the two results together delimit where each ingredient earns its keep. All artifacts are released.

Artifacts. github.com/GHDaru/activelearning (experiments/p1, p2).

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